

SECTION A (25 MARKS)

NO	ANSWER SCHEME	MARKS
1(a)	$\lim_{x \rightarrow -3} \frac{x+3}{x^3+27} = \frac{0}{0}$ $\begin{array}{r} x^2-3x+9 \\ x+3 \overline{) x^3+0x^2+0x+27} \\ \underline{-(x^3+3x^2)} \\ -3x^2+0x \\ \underline{-(-3x^2-9x)} \\ 9x+27 \\ \underline{-(9x+27)} \\ \bullet \bullet \end{array}$ $\lim_{x \rightarrow -3} \frac{x+3}{(x+3)(x^2-3x+9)} = \lim_{x \rightarrow -3} \frac{1}{x^2-3x+9}$ $= \frac{1}{27}$	K1 – Long division K1 – Simplify K1 – Substitute value J1 – Answer
(b)	$\lim_{x \rightarrow \infty} \sqrt{\frac{5x+7}{6x-5}} = \frac{\infty}{\infty}$ $\lim_{x \rightarrow \infty} \sqrt{\frac{\frac{5x+7}{x}}{\frac{6x-5}{x}}} = \lim_{x \rightarrow \infty} \sqrt{\frac{5+\frac{7}{x}}{6-\frac{5}{x}}}$ $= \sqrt{\frac{5+0}{6-0}}$ $= \sqrt{\frac{5}{6}} \text{ or } \frac{\sqrt{30}}{6}$	K1 – Divide by x K1 – Substitute value J1 – Answer
(c)	$\lim_{x \rightarrow 2^-} \frac{x+1}{x^2-4} = \frac{3}{0^-} = -\infty$	K1 J1 – Answer
TOTAL		9 MARKS

NO	ANSWER SCHEME	MARKS
2 a (i)	$y = 8^{7x+6}$ $\frac{dy}{dx} = 8^{7x+6} \frac{d}{dx}(7x+6) \ln 8$ $\frac{dy}{dx} = 7(\ln 8)8^{7x+6}$	K1 J1
a (ii)	$f(x) = \ln\left(\frac{x^2-1}{\sqrt{x-1}}\right)$ $= \ln(x^2-1) - \ln(\sqrt{x-1})$ $= \ln(x^2-1) - \frac{1}{2}\ln(x-1)$ $f'(x) = \frac{1}{x^2-1}(2x) - \frac{1}{2}\left(\frac{1}{x-1}\right)(1)$ $= \frac{2x}{x^2-1} - \frac{1}{2(x-1)}$	K1 (Rule of log) K1 (Differentiate) J1
(b)	$u = x \quad v = \cos x$ $u' = 1 \quad v' = -\sin x$ $\frac{dy}{dx} = \cos 2x(2) - [(\cos x)(1) + (x)(-\sin x)]$ $= 2 \cos 2x - \cos x + x \sin x$ $\text{When } x = \frac{\pi}{3}, \frac{dy}{dx} = 2 \cos\left(2\left(\frac{\pi}{3}\right)\right) - \cos\left(\frac{\pi}{3}\right) + \left(\frac{\pi}{3}\right)\sin\left(\frac{\pi}{3}\right)$ $= -0.5931$	B1 (u' and v') K1 (Differentiate) K1 (Sub) J1
TOTAL		9 MARKS

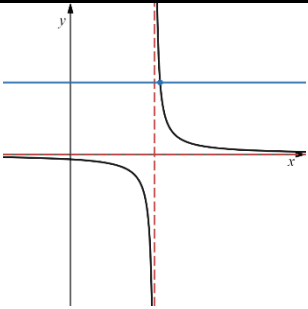
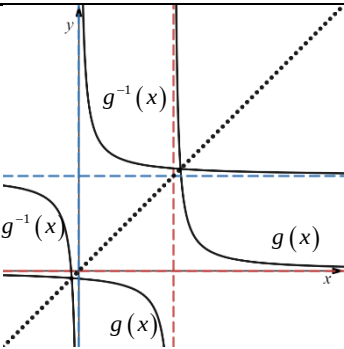
NO	ANSWER SCHEME	MARKS
3 (a)	$f(x) = 27 + 9x - 3x^2 - x^3$ $f'(x) = 9 - 6x - 3x^2$ $f'(x) = 0 \Rightarrow 9 - 6x - 3x^2 = 0$ $x^2 + 2x - 3 = 0$ $(x+3)(x-1) = 0$ $x = -3, x = 1$	K1 – Differentiate
	$\text{When } x = -3, y = 27 + 9(-3) - 3(-3)^2 - (-3)^3 = 0$	K1 – Find x
(b)	$\text{When } x = 1, y = 27 + 9(1) - 3(1)^2 - (1)^3 = 32$ Stationary points = $(-3, 0), (1, 32)$	K1 – Find y
	$f''(x) = -6 - 6x$ $\text{When } x = -3, f''(x) = -6 - 6(-3) = 12 (> 0) \text{ min}$ $\text{When } x = 1, f''(x) = -6 - 6(1) = -12 (< 0) \text{ max}$ $(-3, 0) \text{ is a minimum point, } (1, 32) \text{ is a maximum point}$	J1 J1 – Answer
TOTAL		7 MARKS
SECTION A		25 MARKS

SECTION B (75 MARKS)

NO	ANSWER SCHEME	MARKS
1	$\frac{\overline{(z_1 z_2)}}{13} + \overline{\left(\frac{i^3}{-z_2}\right)}$ $\frac{i^3}{-z_2} = \frac{i^2 i}{-(3+2i)}$ $= \frac{-i}{-3-2i} \times \frac{-3+2i}{-3+2i}$ $= \frac{3i-2i^2}{9-4i^2}$ $= \frac{2+3i}{13}$ $= \frac{2}{13} + \frac{3}{13}i$ $\overline{\left(\frac{i^3}{-z_2}\right)} = \frac{2}{13} - \frac{3}{13}i$ $\frac{\overline{(z_1 z_2)}}{13} + \overline{\left(\frac{i^3}{-z_2}\right)} = \frac{(3+3i)(3+2i)}{13} + \frac{2}{13} - \frac{3}{13}i$ $= \frac{9+6i+9i+6i^2+2-3i}{13}$ $= \frac{5+12i}{13}$ $= \frac{5}{13} + \frac{12}{13}i$	B1 – Conjugate K1 – Simplify J1 B1 – Answer K1 - Simplify J1
TOTAL		6 MARKS

NO	ANSWER SCHEME	MARKS
2 (a)	$x^{\ln x - 2} = e^3$ $\ln x^{\ln x - 2} = \ln e^3$ $(\ln x - 2) \ln x = 3$ $(\ln x)^2 - 2 \ln x - 3 = 0$ Let $u = \ln x$ $u^2 - 2u - 3 = 0$ $(u - 3)(u + 1) = 0$ $u = 3 \quad u = -1$ $\ln x = 3 \quad \ln x = -1$ $x = e^3 \quad x = e^{-1}$ $\therefore x = e^3 \quad x = e^{-1}$	K1 – ln both sides K1 – Attempt to solve K1 – Quadratic eqn J1 – Answer K1 – Find x J1 – Answer
(b)	$x - 1 \leq x^2 + 3x \leq x + 3$ $x - 1 \leq x^2 + 3x \quad \text{and} \quad x^2 + 3x \leq x + 3$ $-x^2 - 2x - 1 \leq 0 \quad x^2 + 2x - 3 \leq 0$ $x^2 + 2x + 1 \geq 0$ $(x + 1)(x + 1) \geq 0 \quad (x - 1)(x + 3) \leq 0$ $\therefore x \in \mathbb{R}$ $\therefore \{x: -3 \leq x \leq 1\}$	B1 – Separate K1 – Quadratic Ineq K1 – Factorise K1 – Method K1 – Intersection J1 – Solution set
TOTAL		12 MARKS

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	 Horizontal Line intersect the graph at only one point, $g(x)$ is a one-to-one function $\frac{1}{2g^{-1}(x) - 5} = x$ $\frac{1}{x} = 2g^{-1}(x) - 5$ $\frac{1}{x} + 5 = 2g^{-1}(x)$ $\frac{\frac{1}{x} + 5}{2} = g^{-1}(x)$ $g^{-1}(x) = \frac{1}{2x} + \frac{5}{2}$	J1 K1 – Substitute K1 – Attempt to solve J1 – Answer
(c)		R1 R1 R1
(d)	$gg^{-1}(x) = \frac{1}{2\left(\frac{1}{2x} + \frac{5}{2}\right) - 5}$ $= \frac{1}{\frac{1}{x} + 5 - 5}$ $= \frac{1}{\frac{1}{x}}$ $= x \quad (\text{shown})$	K1 – Substitute J1
TOTAL		12 MARKS

NO	ANSWER SCHEME	MARKS
5 (a)(i)	$\lim_{x \rightarrow 2^-} x^2 - 4x + a = \lim_{x \rightarrow 2^+} x + 2$ $(2)^2 - 4(2) + a = 2 + 2$ $-4 + a = 4$ $a = 8$ $\lim_{x \rightarrow 2^-} 4 - x = \lim_{x \rightarrow 2^+} \frac{b}{x}$ $4 - 2 = \frac{b}{2}$ $4 = b$	K1 J1 K1 J1 – a and b
(a)(ii)	For $x \geq 2$ $f + g = x + 2 + \frac{4}{x} = \frac{x^2 + 2x + 4}{x}$ For $x < 2$ $f + g = x^2 - 4x + 8 + 4 - x = x^2 - 5x + 12$ $(f + g)(x) = \begin{cases} \frac{x^2 + 2x + 4}{x}, & x \geq 2 \\ x^2 - 5x + 12, & x < 2 \end{cases}$ (i) $f(2) = \frac{(2)^2 + 2(2) + 4}{2} = 6$ (ii) $\lim_{x \rightarrow 2^-} x^2 - 5x + 12$ $\lim_{x \rightarrow 2^+} \frac{x^2 + 2x + 4}{x}$ $= (2)^2 - 5(2) + 12$ $= 6$ $= \frac{(2)^2 + 2(2) + 4}{2}$ $= 6$ $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$ $\therefore \lim_{x \rightarrow 2} f(x)$ exist (iii) $f(2) = \lim_{x \rightarrow 2} f(x)$ $\therefore$ function is continuous at $x = 2$	B1 – $f + g$ K1 – 1 st or 2 nd condition K1 – 3 rd condition J1

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NO	ANSWER SCHEME	MARKS
6 a (i)	$x = 3t - t^{-1} \qquad y = t + 3t^{-1}$ $\frac{dx}{dt} = 3 + t^{-2} = 3 + \frac{1}{t^2} \qquad \frac{dy}{dt} = 1 - 3t^{-2} = 1 - \frac{3}{t^2}$ $= \frac{3t^2 + 1}{t^2} \qquad = \frac{t^2 - 3}{t^2}$ $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ $= \frac{t^2 - 3}{t^2} \cdot \frac{t^2}{3t^2 + 1} = \frac{t^2 - 3}{3t^2 + 1} \text{ (shown)}$	K1 (diff x) K1 (diff y) K1 (chain rule) J1
a (ii)	$u = t^2 - 3 \qquad v = 3t^2 + 1$ $u' = 2t \qquad v' = 6t$ $\frac{d^2y}{dx^2} = \frac{(3t^2 + 1)(2t) - (t^2 - 3)(6t)}{(3t^2 + 1)^2} \cdot \frac{dt}{dx}$ $= \frac{6t^3 + 2t - 6t^3 + 18t}{(3t^2 + 1)^2} \cdot \frac{t^2}{3t^2 + 1}$ $= \frac{20t^3}{(3t^2 + 1)^3}$ $y = \frac{7}{2}$ $t + \frac{3}{t} = \frac{7}{2}$ $2t^2 - 7t + 6 = 0$ $(t - 2)(2t - 3) = 0$ $t = 2, t = \frac{3}{2}$ $\therefore t \in \mathbb{Z}, t = 2$ $\left. \frac{d^2y}{dx^2} \right _{t=2} = \frac{20(2)^3}{(3(2)^2 + 1)^3} = \frac{160}{2197}$	B1 (u' and v') K1 K1 J1 K1 (sub) K1 K1
(b)	$3t^2 + 1 \sqrt[3]{t^2 - 3}$ $\frac{t^2 + 1/3}{-10/3}$ $\frac{dy}{dx} = \frac{1}{3} - \frac{10}{3(3t^2 + 1)} = \frac{1}{3} - \frac{10}{3(3t^2 + 1)}$	J1 J1

		K1 (long division)
		J1
TOTAL		15 MARKS
NO	ANSWER SCHEME	MARKS
7	$V = (8 - 2x)(15 - 2x)(x)$ $= (120 - 16x - 30x + 4x^2)(x)$ $= 120x - 46x^2 + 4x^3$ $\frac{dV}{dx} = 120 - 92x + 12x^2$ $0 = 120 - 92x + 12x^2$ $0 = 30 - 23x + 3x^2$ $0 = (3x - 5)(x - 6)$ $x = \frac{5}{3}, x = 6$ $\frac{d^2V}{dx^2} = -92 + 24x$ When $x = \frac{5}{3}, -92 + 24\left(\frac{5}{3}\right) = -52 (< 0)$ max When $x = 6, -92 + 24(6) = 52 (> 0)$ min $\text{width} = 8 - 2\left(\frac{5}{3}\right) = \frac{14}{3}$ $\text{length} = 15 - 2\left(\frac{5}{3}\right) = \frac{35}{3}$ $\text{height} = \frac{5}{3}$	B1 – Volume K1 – Differentiate K1 – Factorise J1 K1 – 2 nd derivative K1K1 – Determine min/max K1K1 – Find width/length / height J1 – All correct
TOTAL		10 MARKS
SECTION B		75 MARKS

SECTION A(25 MARKS)

BIL	SCHEME	MARKS
1.(a)	$\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 6} = \lim_{x \rightarrow -\infty} \frac{\sqrt{\frac{3x^4}{x^4} + \frac{x}{x^4}}}{\frac{x^2 - 6}{x^2}}$ $= \lim_{x \rightarrow -\infty} \frac{\sqrt{3 + \frac{1}{x^3}}}{1 - \frac{6}{x^2}}$ $= \sqrt{3}$	K1 – Numerator K1 – Denominator K1 J1
1(b)	$\lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$ $= \lim_{x \rightarrow -3} \frac{(2 - \sqrt{x^2 - 5})}{x + 3} \times \frac{(2 + \sqrt{x^2 - 5})}{(2 + \sqrt{x^2 - 5})}$ $= \lim_{x \rightarrow -3} \frac{(4 - (x^2 - 5))}{(x + 3)(2 + \sqrt{x^2 - 5})}$ $= \lim_{x \rightarrow -3} \frac{9 - x^2}{(x + 3)(2 + \sqrt{x^2 - 5})}$ $= \lim_{x \rightarrow -3} \frac{(3 - x)(\cancel{3 + x})}{(\cancel{x + 3})(2 + \sqrt{x^2 - 5})}$ $= \lim_{x \rightarrow -3} \frac{(3 - x)}{(2 + \sqrt{x^2 - 5})}$ $= \frac{(3 - (-3))}{(2 + \sqrt{(-3)^2 - 5})}$ $= \frac{6}{(2 + \sqrt{9 - 5})} = \frac{6}{2 + 2} = \frac{3}{2}$	K1 – Multiply with conjugate K1 – Expand K1 – Simplify K1 – Substitute J1
TOTAL		9 Marks

NO	ANSWER SCHEME	MARKS
2(a)	$y = \sin(x^2 + 1)$ $\frac{dy}{dx} = 2x \cos(x^2 + 1)$ $\frac{d^2 y}{dx^2} = \frac{d}{dx}(2x \cos(x^2 + 1))$ $u = 2x \quad v = \cos(x^2 + 1)$ $u' = 2 \quad v' = -2x \sin(x^2 + 1)$ $\frac{d^2 y}{dx^2} = (2)(\cos(x^2 + 1)) + (2x)(-2x \sin(x^2 + 1))$ $\frac{d^2 y}{dx^2} = 2 \cos(x^2 + 1) - 4x^2 \sin(x^2 + 1)$ <i>multiply both sides by x ;</i> $x \frac{d^2 y}{dx^2} = 2x \cos(x^2 + 1) - 4x^3 \sin(x^2 + 1)$ $x \frac{d^2 y}{dx^2} = \frac{dy}{dx} - 4x^3 y$ $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 4x^3 y = 0 \text{ (shown)}$	K1 B1 K1 K1 J1
2(b)	$e^{3xy} = 11x - 5y$ $e^{3xy} \left(3y + 3x \frac{dy}{dx} \right) = 11 - 5 \frac{dy}{dx}$ $3ye^{3xy} + 3xe^{3xy} \frac{dy}{dx} = 11 - 5 \frac{dy}{dx}$ $(3xe^{3xy} + 5) \frac{dy}{dx} = 11 - 3ye^{3xy}$ $\frac{dy}{dx} = \frac{11 - 3ye^{3xy}}{3xe^{3xy} + 5}$ $u = 3x \quad v = y$ $u' = 3 \quad v' = \frac{dy}{dx}$	K1 (attempt to diff) K1 (collect & factorize) K1 J1
TOTAL		9 Marks

BIL	SCHEME	MARKS
3.	$f(x) = \frac{\ln(x-1)}{x-1}$ $u = \ln(x-1) \quad v = x-1$ $u' = \frac{1}{x-1} \quad v' = 1$ $f'(x) = \frac{(x-1) \cdot \frac{1}{x-1} - \ln(x-1)}{(x-1)^2}$ $= \frac{1 - \ln(x-1)}{(x-1)^2}$ $f'(x) = 0$ $\frac{1 - \ln(x-1)}{(x-1)^2} = 0$ $1 - \ln(x-1) = 0$ $1 = \ln(x-1)$ $e = x-1$ $x = e+1, \quad f(e+1) = \frac{1}{e}$ $f'(x) = \frac{1 - \ln(x-1)}{(x-1)^2}$ $= \frac{1}{(x-1)^2} - \frac{\ln(x-1)}{(x-1)^2} = (x-1)^{-2} - \frac{\ln(x-1)}{(x-1)^2}$ $u = \ln(x-1) \quad v = (x-1)^2$ $u' = \frac{1}{x-1} \quad v' = 2(x-1)$ $f''(x) = \frac{-2}{(x-1)^3} - \frac{(x-1)^2 \cdot \frac{1}{x-1} - 2(x-1)\ln(x-1)}{(x-1)^4}$ $= -\frac{2}{(x-1)^3} - \frac{x-1 - 2(x-1)\ln(x-1)}{(x-1)^4}$	K1 u, v, u', v' K1 quotient rule K1 equate to zero K1 Find x K1 differentiate

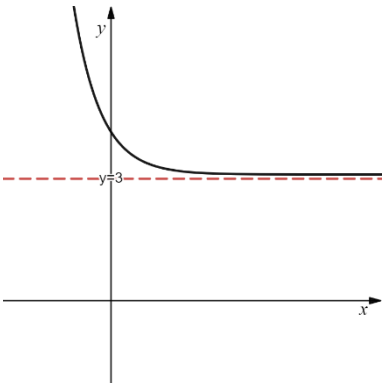
	when $x = e + 1$, $f''(e+1) = -\frac{2}{(e+1-1)^3} - \frac{e+1-1-2(e+1-1)\ln(e+1-1)}{(e+1-1)^4}$ $= -0.0498$ $< 0 \quad (\text{max})$ So, there is only one maximum point which is $\left(e+1, \frac{1}{e}\right)$	K1 Determine min/max J1 – conclusion
TOTAL		7 Marks
SECTION A		25 Marks

SECTION B(75 MARKS)

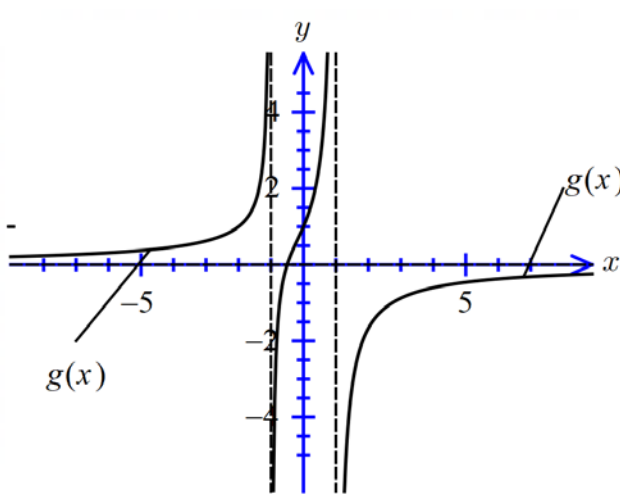
BIL	SCHEME	MARKS
1.	$\frac{p+qi}{3i} = 3 + \sqrt{-16} - i^3$ $p+qi = 3i(3+4i+i) \quad \sqrt{-16} = 4i$ $= 9i + 15i^2 \quad i^3 = -i$ $p+qi = -15 + 9i$ Compare, $\therefore p = -15, \quad q = 9.$ Alternative : $\frac{(p+qi)}{3i} = (3 + \sqrt{-16} - i^3)$ $\frac{(p+qi)}{3i} \times \frac{-3i}{-3i} = (3 + 4i - i^2(i))$ $\frac{q}{3} - \frac{p}{3}i = 3 + 5i$ $\frac{q}{3} = 3 \quad \frac{-p}{3} = 5$ $q = 9 \quad p = -15$	B1 B1 K1 - simplify K1 - compare J1 J1
TOTAL		6 Marks

2(a)	$\sqrt{7-3\sqrt{5}} = \sqrt{x} - \sqrt{y}$ $\left(\sqrt{7-3\sqrt{5}}\right)^2 = \left(\sqrt{x} - \sqrt{y}\right)^2$ $7-3\sqrt{5} = x - 2\sqrt{xy} + y$ Compare $x + y = 7$ $x = 7 - y \text{ -----(1)}$ $3\sqrt{5} = 2\sqrt{xy}$ $\left(3\sqrt{5}\right)^2 = \left(2\sqrt{xy}\right)^2$ $45 = 4xy \text{ -----(2)}$ Subs (1) into (2) $4(7-y)y = 45$ $28y - 4y^2 = 45$ $4y^2 - 28y + 45 = 0$ $(2y-5)(2y-9) = 0$ $y = \frac{5}{2} \quad \text{or} \quad y = \frac{9}{2}$ When $y = \frac{5}{2}$, $x = \frac{9}{2}$ When $y = \frac{9}{2}$, $x = \frac{5}{2}$ Since $\sqrt{x} - \sqrt{y} > 0$ $x > y$ So, $x = \frac{9}{2}$ and $y = \frac{5}{2}$	K1 –square both sides K1 - compare K1 – Attempt to solve K1 – Find y K1 – Find x J1 - Answer
2(b)	$\frac{-4-x}{x-3} \geq x+4, \quad x \neq 3.$ $\frac{-4-x}{x-3} - (x+4) \geq 0$ $\frac{-4-x-(x+4)(x-3)}{x-3} \geq 0$ $\frac{-4-x-(x^2+x-12)}{x-3} \geq 0$ $\frac{-x^2-2x+8}{x-3} \geq 0$	K1 – equate denominator K1 – simplify

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4(a)	 $D_f = \mathbb{R}, R_f = (3, \infty)$	K1 – Method B1 B1
4(b)	$gf(x) = \frac{1}{2} \ln(e^{-2x} + 2)$ $g(e^{-2x} + 3) = \frac{1}{2} \ln(e^{-2x} + 2)$ Let $e^{-2x} + 3 = u$ $e^{-2x} = u - 3$ $-2x = \ln(u - 3)$ $x = \frac{\ln(u - 3)}{-2}$ $g(u) = \frac{1}{2} \ln \left(e^{-2 \left(\frac{\ln(u-3)}{-2} \right)} + 2 \right)$ $= \frac{1}{2} \ln(e^{\ln(u-3)} + 2)$ $= \frac{1}{2} \ln(u - 3 + 2)$ $= \frac{1}{2} \ln(u - 1)$ $\therefore g(x) = \frac{1}{2} \ln(x - 1)$	K1 J1 K1 – Substitute K1 – Simplify J1
4(c)	$\frac{1}{2} \ln(g^{-1}(x) - 1) = x$ $\ln(g^{-1}(x) - 1) = 2x$ $g^{-1}(x) - 1 = e^{2x}$ $g^{-1}(x) = e^{2x} + 1$	K1 – Substitute K1 – Simplify J1

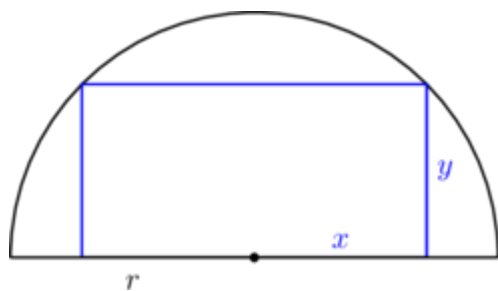
		R1 R1 R1
	TOTAL	14 Marks
5.(a)	$2x + 1 = 0$ $x = -\frac{1}{2}$ $f(x) = \lim_{x \rightarrow -\frac{1}{2}^+} \frac{1-x^2}{2x+1} = +\infty$ $f(x) = \lim_{x \rightarrow -\frac{1}{2}^-} \frac{1-x^2}{2x+1} = -\infty$ There is one vertical asymptote and the equation is $x = -\frac{1}{2}$	K1 – =0 K1 – Both J1 – VA
5(b)	$g(x) = \frac{1}{\frac{1-x^2}{2x+1}} = \frac{2x+1}{1-x^2}$ $1-x^2 = 0$ $x^2 - 1 = 0$ $(x-1)(x+1) = 0$ $x = -1 \quad \text{or} \quad x = 1$ $g(x) = \lim_{x \rightarrow -1^+} \frac{2x+1}{1-x^2} = \lim_{x \rightarrow -1^+} \frac{2x+1}{(1-x)(1+x)} = -\infty$ $g(x) = \lim_{x \rightarrow -1^-} \frac{2x+1}{1-x^2} = \lim_{x \rightarrow -1^-} \frac{2x+1}{(1-x)(1+x)} = +\infty$ $g(x) = \lim_{x \rightarrow 1^+} \frac{2x+1}{1-x^2} = \lim_{x \rightarrow 1^+} \frac{2x+1}{(1-x)(1+x)} = -\infty$ $g(x) = \lim_{x \rightarrow 1^-} \frac{2x+1}{1-x^2} = \lim_{x \rightarrow 1^-} \frac{2x+1}{(1-x)(1+x)} = +\infty$ There vertical asymptotes are $x = -1$ and $x = 1$.	K1 K1 K1 J1 K1 K1

5(c)	$\begin{aligned}\lim_{x \rightarrow +\infty} g(x) &= \lim_{x \rightarrow +\infty} \frac{2x+1}{1-x^2} \\ &= \lim_{x \rightarrow +\infty} \frac{\frac{2x}{x^2} + \frac{1}{x^2}}{\frac{1}{x^2} - \frac{x^2}{x^2}} \\ &= \lim_{x \rightarrow +\infty} \frac{\frac{2}{x} + \frac{1}{x^2}}{\frac{1}{x^2} - 1} \\ &= \frac{0+0}{0-1} \\ &= 0\end{aligned}$ $\begin{aligned}\lim_{x \rightarrow -\infty} g(x) &= \lim_{x \rightarrow -\infty} \frac{2x+1}{1-x^2} \\ &= \lim_{x \rightarrow -\infty} \frac{\frac{2x}{x^2} + \frac{1}{x^2}}{\frac{1}{x^2} - \frac{x^2}{x^2}} \\ &= \lim_{x \rightarrow -\infty} \frac{\frac{2}{x} + \frac{1}{x^2}}{\frac{1}{x^2} - 1} \\ &= \frac{0+0}{0-1} \\ &= 0\end{aligned}$ The horizontal asymptote is $y = 0$. 	J1 – HA R1 R1 R1
TOTAL		14 Marks

NO	ANSWER SCHEME	MARKS
6 (a)	$y = e^x \ln(1+x)$ $u = e^x \quad v = \ln(1+x)$ $u' = e^x \quad v' = \frac{1}{1+x}$ $\frac{dy}{dx} = \ln(1+x)(e^x) + e^x \left(\frac{1}{1+x} \right)$ $u = e^x \quad v = \frac{1}{1+x}$ $u' = e^x \quad v' = -\frac{1}{(1+x)^2}$ $\frac{d^2y}{dx^2} = \ln(1+x)(e^x) + e^x \left(\frac{1}{1+x} \right) + \left(\frac{1}{1+x} \right) e^x - \frac{e^x}{(1+x)^2}$ $\frac{d^2y}{dx^2} = \frac{dy}{dx} + \frac{(1+x)(e^x) - e^x}{(1+x)^2}$ $\frac{d^2y}{dx^2} - \frac{dy}{dx} = \frac{e^x + xe^x - e^x}{(1+x)^2}$ $\frac{d^2y}{dx^2} - \frac{dy}{dx} = \frac{xe^x}{(1+x)^2}$ $(x+1)^2 \left(\frac{d^2y}{dx^2} - \frac{dy}{dx} \right) = xe^x \text{ (shown)}$ Alternative: $y = e^x \ln(1+x)$ $\frac{dy}{dx} = e^x \left(\frac{1}{1+x} \right) + \ln(1+x)(e^x)$ $\frac{dy}{dx} = \left(\frac{e^x}{1+x} \right) + y$ $\frac{d^2y}{dx^2} = \frac{(1+x)(e^x) - e^x}{(1+x)^2} + \frac{dy}{dx}$ $\frac{d^2y}{dx^2} - \frac{dy}{dx} = \frac{e^x + xe^x - e^x}{(1+x)^2}$ $(x+1)^2 \left(\frac{d^2y}{dx^2} - \frac{dy}{dx} \right) = xe^x \text{ (shown)}$	K1 J1 J1 K1 K1 J1

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7(a)



$$x^2 + y^2 = 100$$

$$y = \sqrt{100 - x^2}$$

$$\text{Area of rectangle: } A = 2xy = 2x\sqrt{100 - x^2}$$

$$A' = 2\sqrt{100 - x^2} - \frac{2x^2}{\sqrt{100 - x^2}}$$

$$0 = 2\sqrt{100 - x^2} - \frac{2x^2}{\sqrt{100 - x^2}}$$

$$\frac{2x^2}{\sqrt{100 - x^2}} = 2\sqrt{100 - x^2}$$

$$2x^2 = 2(100 - x^2)$$

$$4x^2 = 200$$

$$x = \sqrt{50}, \quad y = \sqrt{50}$$

$$A' = 2\sqrt{100 - x^2} - \frac{2x^2}{\sqrt{100 - x^2}}$$

$$A' = 2(100 - x^2)^{-\frac{1}{2}} [100 - x^2 - x^2]$$

$$A' = 2(100 - x^2)^{-\frac{1}{2}} [100 - 2x^2]$$

$$u = 2(100 - x^2)^{-\frac{1}{2}} \quad v = 100 - 2x^2$$

$$u' = -\frac{1}{2}(2)(100 - x^2)^{-\frac{3}{2}}(-2x) \quad v' = -4x$$

$$= 2x(100 - x^2)^{-\frac{3}{2}}$$

B1 – Area

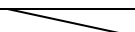
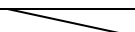
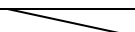
K1 –
differentiate
and equate to
zero.

K1 – Find x
and y

	$A'' = (100 - 2x^2)(2x)(100 - x^2)^{-\frac{3}{2}} + 2(100 - x^2)^{-\frac{1}{2}}(-4x)$ $x = \sqrt{50},$ $A'' = (100 - 2\sqrt{50}^2)(2\sqrt{50})(100 - \sqrt{50}^2)^{-\frac{3}{2}} - 8\sqrt{50}(100 - \sqrt{50}^2)^{-\frac{1}{2}}$ $= -8 \quad (< 0, \text{max})$ When $x = \sqrt{50}, y = \sqrt{50}$ The maximum area of rectangle, $A = 2\sqrt{50}\sqrt{100 - (\sqrt{50})^2}$ $= 100\text{cm}^2.$	K1 – determine min/max J1 – Answer
7(b)	$r = h \tan 30^\circ \quad \frac{dV}{dt} = -6$ $= \frac{h}{\sqrt{3}}$ $V = \frac{1}{3}\pi r^2 h \quad 24\pi = \frac{1}{9}\pi h^3$ $= \frac{1}{3}\pi \left(\frac{h}{\sqrt{3}}\right)^2 h \quad h = 6$ $= \frac{1}{9}\pi h^3$ $\frac{dV}{dh} = \frac{\pi h^2}{3}$ $\frac{dV}{dt} = \frac{\pi h^2}{3} \cdot \frac{dh}{dt}$ $-6 = \frac{\pi h^2}{3} \cdot \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{-18}{\pi h^2}$ $= \frac{-18}{\pi(6)^2}$ $h = -\frac{1}{2\pi}\text{cms}^{-1}$	K1 – Find r in terms of h K1 – Volume K1 – Find h K1 – Find $\frac{dh}{dt}$ J1 – Answer
	TOTAL	10 Marks
	SECTION B	75 Marks

SECTION A (25 MARKS)

BIL	SCHEME	MARKS
1(a)	$ x - 2 = \begin{cases} x - 2, & x \geq 2 \\ -(x - 2), & x < 2 \end{cases}$ $\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{ x - 2 } = \lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 2)}{-(x - 2)}$ $= \lim_{x \rightarrow 2^-} -(x + 2)$ $= -4$	K1 factorize K1 $-(x - 2)$ K1 Simplify J1
(b)	$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 3x}}{5x + 1} = \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{2x^2}{x^2} + \frac{3x}{x^2}}}{\frac{5x}{x} + \frac{1}{x}}$ $= \lim_{x \rightarrow -\infty} \frac{-\sqrt{2 + \frac{3}{x}}}{5 + \frac{1}{x}}$ $= \frac{-\sqrt{2 + 0}}{5 + 0}$ $= \frac{-\sqrt{2}}{5}$	K1 Divide by $-x$ (BOTH) K1 $\sqrt{\frac{2x^2}{x^2} + \frac{3x}{x^2}}$ K1 Simplify K1✓ Substitute J1
TOTAL		9 Marks
2 (a)	$y = x^{\sec x}$ $\ln y = \sec x \ln x$ $\frac{1}{y} \frac{dy}{dx} = \frac{\sec x}{x} + (\ln x) \sec x \tan x$ $\frac{dy}{dx} = y \sec x \left(\frac{1}{x} + \tan x \ln x \right)$ $= x^{\sec x} \sec x \left(\frac{1}{x} + \tan x \ln x \right)$	K1 ln both sides & use prop K1 Diff LHS correctly K1 Diff RHS correctly K1✓ Simplify J1

(b)	$y = \cot \sqrt{4x^2 + 1}$ $= \cot(4x^2 + 1)^{\frac{1}{2}}$ $\frac{dy}{dx} = -\operatorname{cosec}^2(4x^2 + 1)^{\frac{1}{2}} \frac{d}{dx}[(4x^2 + 1)^{\frac{1}{2}}]$ $= -\operatorname{cosec}^2(4x^2 + 1)^{\frac{1}{2}} \left[\frac{1}{2}(4x^2 + 1)^{-\frac{1}{2}}(8x) \right]$ $= -\frac{4x}{\sqrt{4x^2 + 1}} \operatorname{cosec}^2 \sqrt{(4x^2 + 1)}$	K1 $-\operatorname{cosec}^2(4x^2 + 1)^{\frac{1}{2}}$ K1 $\frac{1}{2}(4x^2 + 1)^{-\frac{1}{2}}(8x)$ K1 All correct J1												
TOTAL		9 Marks												
3	$f(x) = x^3 - \frac{3}{2}x^2 - 18x + 1$ $f'(x) = 3x^2 - 3x - 18$ $f'(x) = 0$ $3x^2 - 3x - 18 = 0$ $x^2 - x - 6 = 0$ $(x - 3)(x + 2) = 0$ $x = 3, x = -2$ $f(-2) = (-2)^3 - \frac{3}{2}(-2)^2 - 18(-2) + 1 = 23$ $f(3) = (3)^3 - \frac{3}{2}(3)^2 - 18(3) + 1 = -\frac{79}{2}$ <table border="1"><tr><td>Interval</td><td>$(-\infty, -2)$</td><td>$(-2, 3)$</td><td>$(3, \infty)$</td></tr><tr><td>Sign of $f'(x)$</td><td>+</td><td>-</td><td>+</td></tr><tr><td>Conclusion</td><td></td><td></td><td></td></tr></table> $\therefore (-2, 23) \text{ is maximum point}$ $\left(3, -\frac{79}{2} \right) \text{ is minimum point}$	Interval	$(-\infty, -2)$	$(-2, 3)$	$(3, \infty)$	Sign of $f'(x)$	+	-	+	Conclusion				K1 Differentiate correctly K1 Equate to 0 J1 Both K1 Find y K1✓ First derivative test J1 Max point J1 Min point
Interval	$(-\infty, -2)$	$(-2, 3)$	$(3, \infty)$											
Sign of $f'(x)$	+	-	+											
Conclusion														
TOTAL		7 Marks												
SECTION A		25 Marks												

SECTION B (75 MARKS)

BIL	SCHEME	MARKS
1	$\bar{z} - \frac{1}{z} = \frac{8}{5} - \frac{6}{5}i$ $\left \bar{z} - \frac{1}{z} \right = \sqrt{\left(\frac{8}{5}\right)^2 + \left(-\frac{6}{5}\right)^2} = 2$ $\alpha + \beta i = \left \bar{z} - \frac{1}{z} \right \left(\bar{z} - \frac{1}{z} \right)^2$ $= 2 \left(\frac{8}{5} - \frac{6}{5}i \right)^2$ $= 2 \left(\frac{8}{5} - \frac{6}{5}i \right) \left(\frac{8}{5} - \frac{6}{5}i \right)$ $= 2 \left(\frac{64}{25} - \frac{96}{25}i + \frac{36}{25}i^2 \right)$ $= 2 \left(\frac{28}{25} - \frac{96}{25}i \right)$ $= \frac{56}{25} - \frac{192}{25}i$	K1 J1 K1✓ K1 Correct expansion K1✓ Simplify $i^2 = -1$ J1
TOTAL		6 Marks
2(a)	$2 + \log_2 x = 15 \log_x 2$ $2 + \log_2 x = 15 \left(\frac{\log_2 2}{\log_2 x} \right)$ Let $u = \log_2 x$ $2 + u = \frac{15}{u}$ $u^2 + 2u - 15 = 0$ $(u - 3)(u + 5) = 0$ $u = 3, u = -5$ $\log_2 x = 3 \qquad \log_2 x = -5$ $x = 8 \qquad x = \frac{1}{32}$ $\therefore x = 8, x = \frac{1}{32}$	K1 Change base K1 Quadratic equation J1 K1✓ Find x J1 Answer

BIL	SCHEME	MARKS
2(b)	$\left x - \frac{2}{x} \right \leq 1$ $x - \frac{2}{x} \leq 1$ $x - \frac{2}{x} - 1 \leq 0$ $\frac{x^2 - x - 2}{x} \leq 0$ $\frac{(x-2)(x+1)}{x} \leq 0$ $x - \frac{2}{x} \geq -1$ $x - \frac{2}{x} + 1 \geq 0$ $\frac{x^2 + x - 2}{x} \geq 0$ $\frac{(x+2)(x-1)}{x} \geq 0$ $(-\infty, -1] \cup (0, 2]$ $[-2, 0) \cup [1, \infty)$ $\therefore [-2, -1] \cup [1, 2]$	B1 Separate K1 Equate denominator K1 Simplify K1✓ Method J1 Answer K1✓ Intersection J1 Answer
	TOTAL	12 Marks

BIL	SCHEME	MARKS
3(a)	$f(x) = 3x + 8$ $h(x) = 3x^2 - 12x + 2$ $(f \circ g)(x) = h(x)$ $3g(x) + 8 = 3x^2 - 12x + 2$ $g(x) = \frac{3x^2 - 12x - 6}{3}$ $g(x) = x^2 - 4x - 2$ $g(x) = (x - 2)^2 - 4 - 2$ $g(x) = (x - 2)^2 - 6$	K1 Substitute (As it is) K1 Find $g(x)$ K1 Simplify K1✓ CTS J1
(b)	$D_g = (-\infty, \infty)$ $R_g = [-6, \infty)$	J1 J1
(c)		R1 Shape R1 Minimum point & y-intercept R1 All correct & label
TOTAL		10 MARKS

BIL	SCHEME	MARKS
4(a)	$f(x) = \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$ $= \frac{1}{1 + \frac{x+1}{x}}$ $= \frac{1}{1 + \frac{x}{x+1}}$ $= \frac{1}{\frac{x+1+x}{x+1}}$ $= \frac{1}{\frac{2x+1}{x+1}}$ $= \frac{x+1}{2x+1}, x \neq -\frac{1}{2}$ $f\left(\frac{1}{2}\right) = \frac{\frac{1}{2} + 1}{2\left(\frac{1}{2}\right) + 1}$ $= \frac{3}{4}$	K1 K1 J1 K1 J1
(b)	Denominator of $f(x) \neq 0$: $x \neq 0 \qquad x+1 \neq 0 \qquad 2x+1 \neq 0$ $\qquad \qquad x \neq -1 \qquad \qquad x \neq -\frac{1}{2}$	B1 B1 B1
TOTAL		8 Marks

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BIL	SCHEME	MARKS
	$\lim_{x \rightarrow +\infty} \left(\frac{x}{x-1} \right) = \lim_{x \rightarrow +\infty} \left(\frac{\frac{x}{x}}{\frac{x-1}{x}} \right)$ $= \lim_{x \rightarrow +\infty} \left(\frac{1}{1 - \frac{1}{x}} \right)$ $= \frac{1}{1-0} = 1$	K1 Divide by x J1
	$\lim_{x \rightarrow -\infty} \left(\frac{x}{-x-1} \right) = \lim_{x \rightarrow -\infty} \left(\frac{\frac{x}{x}}{\frac{-x-1}{x}} \right)$ $= \lim_{x \rightarrow -\infty} \left(\frac{1}{-1 - \frac{1}{x}} \right)$ $= \frac{1}{-1-0} = -1$	K1 Divide by x J1
	$\therefore y = 1$ and $y = -1$ are the horizontal asymptotes.	J1
TOTAL		14 Marks

BIL	SCHEME	MARKS
6(a)	$x^2 - 3y^2 = ae^{y-2x} + by - 6$ At (1,2): $(1)^2 - 3(2)^2 = ae^{2-2(1)} + b(2) - 6$ $1 - 12 = a + 2b - 6$ $a + 2b = -5 \quad \dots (1)$ $\frac{dy}{dx} = 1:$ $x^2 - 3y^2 = ae^{y-2x} + by - 6$ $2x - 6y \frac{dy}{dx} = ae^{y-2x} \frac{d}{dx}(y - 2x) + b \frac{dy}{dx}$ $2x - 6y \frac{dy}{dx} = ae^{y-2x} \left(\frac{dy}{dx} - 2 \right) + b \frac{dy}{dx}$ $2(1) - 6(2)(1) = ae^{(2)-(2(1))} (1 - 2) + b(1)$ $-10 = -a + b$ $a - b = 10 \quad \dots (2)$ $(1) - (2):$ $3b = -15$ $b = -5$ $a + 5 = 10$ $a = 5$ $\therefore a = 5, b = -5$	K1 Substitute (1,2) J1 K1 Diff LHS correctly K1 Diff RHS correctly J1 All correct K1✓ Substitute (1,2) and $\frac{dy}{dx} = 1$ J1 K1✓ Attempt to solve J1 Both

BIL	SCHEME	MARKS
6 (b)	$2x - 6y \frac{dy}{dx} = 5e^{y-2x} \left(\frac{dy}{dx} - 2 \right) - 5 \frac{dy}{dx}$ Diff LHS : $2 - \left[6y \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 6 \frac{dy}{dx} \right]$ Diff RHS : $5e^{y-2x} \left(\frac{d^2y}{dx^2} \right) + \left(\frac{dy}{dx} - 2 \right) (5e^{y-2x}) \left(1 \frac{dy}{dx} - 2 \right) - 5 \frac{d^2y}{dx^2}$ At (1, 2), $\frac{dy}{dx} = 1$: $2 - \left[6(2) \frac{d^2y}{dx^2} + 6(1)(1) \right] = 5e^0 \left(\frac{d^2y}{dx^2} \right) + (1-2)(5e^0)(1-2) - 5 \frac{d^2y}{dx^2}$ $2 - 12 \frac{d^2y}{dx^2} - 6 = 5 \frac{d^2y}{dx^2} + 5 - 5 \frac{d^2y}{dx^2}$ $12 \frac{d^2y}{dx^2} = -9$ $\frac{d^2y}{dx^2} = -\frac{3}{4}$	K1 Diff LHS K1 Diff RHS J1 All correct K1✓ Substitute K1✓ J1
TOTAL		15 Marks

BIL	SCHEME	MARKS
7(a)	Perimeter = 8 $2y + 4x + (\pi)(2x) = 8$ $2y + 4x + 2\pi x = 8$ $2y = 8 - 4x - 2\pi x$ $y = 4 - 2x - \pi x$ Surface area : $A = \frac{1}{2}\pi(2x)^2 + (4x)y$ $A = 2\pi x^2 + 4xy$ $A = 2\pi x^2 + 4x(4 - 2x - \pi x)$ $A = (-2\pi - 8)x^2 + 16x$	K1 Correct formula (As it is) J1 K1 Correct formula & Sub HIS y J1 Answer
7(b)	$\frac{dA}{dx} = (-2\pi - 8)(2x) + 16$ $\frac{dA}{dx} = 0;$ $(-2\pi - 8)(2x) + 16 = 0$ $x = 0.5601 \text{ m}$ $\frac{d^2A}{dx^2} = (-2\pi - 8)(2)$ $= -28.57 < 0 \text{ (Maximum)}$ Length = $4x$ $= 4(0.5601)$ $= 2.2404 \text{ m}$	K1✓ Differentiate K1✓ Equate to zero J1 K1✓ Find $\frac{d^2A}{dx^2}$ and test value K1✓ J1
TOTAL		10 Marks
SECTION B		75 Marks

SECTION A (25 MARKS)

BIL	SCHEME	MARKS
1.(a)	$\lim_{x \rightarrow 0} \frac{mx + 3x^2}{4x - 8x^2} = \lim_{x \rightarrow 0} \frac{x(m + 3x)}{x(4 - 8x)}$ $= \lim_{x \rightarrow 0} \frac{m + 3x}{4 - 8x}$ $= \frac{m + 3(0)}{4 - 8(0)}$ $= \frac{m}{4}$ $\frac{m}{4} = 3 \rightarrow m = 12$	K1 (factorize) K1 (substitute) J1
1.(b)(i)	$\lim_{x \rightarrow 0} \frac{\sqrt{3-x} - \sqrt{3}}{x} = \lim_{x \rightarrow 0} \frac{x}{\sqrt{3-x} - \sqrt{3}} \times \frac{\sqrt{3-x} + \sqrt{3}}{\sqrt{3-x} + \sqrt{3}}$ $= \lim_{x \rightarrow 0} \frac{x(\sqrt{3-x} + \sqrt{3})}{(3-x) - 3}$ $= \lim_{x \rightarrow 0} \frac{x(\sqrt{3-x} + \sqrt{3})}{-x}$ $= -(\sqrt{3-0} + \sqrt{3})$ $= -2\sqrt{3}$	K1 (conjugate) K1 (substitute) J1
1.(b)(ii)	$ 3-x = \begin{cases} 3-x & x \leq 3 \\ x-3 & x > 3 \end{cases}$ $\lim_{x \rightarrow 3^+} \frac{x^2 - 5x + 6}{ 3-x } = \lim_{x \rightarrow 3^+} \frac{x^2 - 5x + 6}{x-3}$ $= \lim_{x \rightarrow 3^+} \frac{(x-2)(x-3)}{x-3}$ $= \lim_{x \rightarrow 3^+} (x-2)$ $= 3-2$ $= 1$	K1 (choose correct function) K1 (correct factorization) J1
	TOTAL	9 Marks

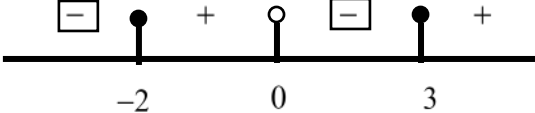
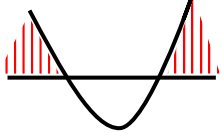
2.	$f(x) = \frac{3x+1}{(x-1)(x^2+1)}$ $\frac{3x+1}{(x-1)(x^2+1)} = \frac{A}{(x-1)} + \frac{(Bx+C)}{(x^2+1)}$ $3x+1 = A(x^2+1) + (Bx+C)(x-1)$ $3x+1 = (A+B)x^2 + (-B+C)x + (A-C)$ sub $x=1$, $4 = 2A$ $A = 2$ comparing coefficients $0 = A+B \qquad 1 = A-C$ $B = -2 \qquad C = 1$ $\frac{3x+1}{(x-1)(x^2+1)} = \frac{2}{(x-1)} + \frac{1-2x}{(x^2+1)}$ $f(x) = \frac{2}{(x-1)} + \frac{1-2x}{(x^2+1)}$ $f(x) = 2(x-1)^{-1} + \frac{1-2x}{(x^2+1)}$ $u = 1-2x \qquad v = x^2+1$ $u' = -2 \qquad v' = 2x$ $f'(x) = 2[(-1)(x-1)^{-2}] + \frac{(x^2+1)(-2) - (1-2x)(2x)}{(x^2+1)^2}$ $= \frac{-2}{(x-1)^2} + \frac{2x^2-2x-2}{(x^2+1)^2}$ $f'(3) = \frac{-2}{(3-1)^2} + \frac{2(3)^2-2(3)-2}{((3)^2+1)^2}$ $= -\frac{2}{5}$	B1 K1 K1 (attempt to solve) J1 K1 J1 K1(sub) J1
	TOTAL	9 Marks

3.	when $x = -1, y = -1$ $-1 = \frac{p(-1) + r}{-1(-1-3)}$ $-1 = \frac{-p + r}{4}$ $-p + r = -4 \quad \text{--- (1)}$ $f'(x) = \frac{(x^2 - 3x)p - (px + r)(2x - 3)}{(x^2 - 3x)^2}$ $= \frac{px^2 - 3px - 2px^2 + 3px - 2rx + 3r}{(x^2 - 3x)^2}$ $= \frac{-px^2 - 2rx + 3r}{(x^2 - 3x)^2}$ At $(-1, -1)$, $f'(x) = 0$ $\frac{-p(-1)^2 - 2r(-1) + 3r}{[(-1)^2 - 3(-1)]^2} = 0$ $-p + 2r + 3r = 0$ $p = 5r \quad \text{--- (2)}$ Substitute (2) into (1) $-(5r) + r = -4$ $r = 1$ $p = 5$ when $f'(x) = 0$, $\frac{-5x^2 - 2x + 3}{(x^2 - 3x)^2} = 0$ $5x^2 + 2x - 3 = 0$ $(5x - 3)(x + 1) = 0$ $x = \frac{3}{5}, x = -1$	K1 K1 - differentiate K1 J1 - (both answer) K1 K1
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	$\text{when } x = \frac{3}{5},$ $f\left(\frac{3}{5}\right) = \frac{5\left(\frac{3}{5}\right) + 1}{\frac{3}{5}\left(\frac{3}{5} - 3\right)} = -\frac{25}{9}$ $\therefore H = \left(\frac{3}{5}, -\frac{25}{9}\right)$	J1
	TOTAL	7 Marks
	SECTION A	25 marks

SECTION B (75 MARKS)

BIL	SCHEME	MARKS
1.(a)	$\sqrt{8+6i} = a+bi$ $8+6i = a^2 + 2abi + b^2i^2$ $8+6i = a^2 - b^2 + 2abi$ $a^2 - b^2 = 8 \qquad 2ab = 6$ $\left(\frac{3}{b}\right)^2 - b^2 = 8 \qquad a = \frac{3}{b}$ $\frac{9}{b^2} - b^2 = 8$ $-b^4 - 8b^2 + 9 = 0$ $b^4 + 8b^2 - 9 = 0$ $(b^2 - 1)(b^2 + 9) = 0$ $b^2 = 1 \qquad b^2 = -9$ $b = \pm 1 \qquad \text{undefined}$ $b = 1, \quad a = \frac{3}{1} = 3$ $b = -1, \quad a = \frac{3}{-1} = -3$ $z = 1 + 3i, \quad z = -1 - 3i$	K1 (expand) K1 (compare) K1 (quadratic) J1 (correct both) K1 (substitute) J1

TOTAL		6 Marks
2(a)	$\log_3 x - 1 \leq 6 \log_x 3$ $\log_3 x - 1 - 6 \log_x 3 \leq 0$ $\log_3 x - 1 - \frac{6}{\log_3 x} \leq 0$ Let $\log_3 x = y$ $y - 1 - \frac{6}{y} \leq 0$ $\frac{y^2 - y - 6}{y} \leq 0$ $\frac{(y-3)(y+2)}{y} \leq 0$ CN: $y = 3, y = -2, y = 0$  $y \leq -2$ $0 < y \leq 3$ $\log_3 x \leq -2$ $x \leq 3^{-2}$ $x \leq \frac{1}{9}$ $0 < \log_3 x \leq 3$ $3^0 < x \leq 3^3$ $1 < x \leq 27$ $\therefore \left\{ x : x \leq \frac{1}{9} \text{ or } 1 < x \leq 27 \right\}$	K1 – change base K1 – equate denominator K1 – factorize K1 – method K1 – correct region K1 – find interval of x J1
2(b)	$ x-1 \leq 2 x+3 $ $\sqrt{(x-1)^2} \leq 2\sqrt{(x+3)^2}$ $(x-1)^2 \leq 4(x+3)^2$ $x^2 - 2x + 1 \leq 4(x^2 + 6x + 9)$ $3x^2 + 26x + 35 \geq 0$ $(3x+5)(x+7) \geq 0$ CN: $x = -\frac{5}{3}, x = -7$ 	K1 – square both sides K1 – expand correctly K1 – factor K1 – method

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4.	$fg(x) = 2x$ $f(e^{2x} - 3) = 2x$ $\text{let } u = e^{2x} - 3$ $2x = \ln(u + 3)$ $x = \frac{1}{2} \ln(u + 3)$ $f(u) = 2 \left[\frac{1}{2} \ln(u + 3) \right]$ $f(u) = \ln(u + 3)$ $\therefore f(x) = \ln(x + 3)$ $f \circ f^{-1}(x) = x$ $\ln(f^{-1}(x) + 3) = x$ $f^{-1}(x) + 3 = e^x$ $f^{-1}(x) = e^x - 3$ $D_{f^{-1}} = \mathfrak{R}$ $R_{f^{-1}} = (-3, \infty)$	K1 (substitute) K1 K1 (substitute) J1 K1 J1 B1 B1
TOTAL		8 Marks
5.(a)(i)	$\lim_{x \rightarrow -2^-} \frac{1+x}{\sqrt{(x+2)(x-2)}} = \frac{-1}{0^+} = -\infty$ $\lim_{x \rightarrow -2^-} \frac{1+x}{\sqrt{(x+2)(x-2)}} = \text{undefined}$ $\lim_{x \rightarrow 2^+} \frac{1+x}{\sqrt{(x+2)(x-2)}} = \frac{3}{0^+} = +\infty$ $\lim_{x \rightarrow 2^-} \frac{1+x}{\sqrt{(x+2)(x-2)}} = \text{undefined}$ The vertical asymptotes are $x=2$ and $x=-2$.	K1 K1 J1
5.(a)(ii)	$\lim_{x \rightarrow +\infty} \frac{\frac{1}{x} + \frac{x}{x}}{\sqrt{\frac{x^2}{x^2} - \frac{4}{x^2}}} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x} + 1}{\sqrt{1 - \frac{4}{x^2}}} = \frac{0+1}{\sqrt{1-0}} = 1$ $\lim_{x \rightarrow -\infty} \frac{\frac{1}{x} + \frac{x}{x}}{-\sqrt{\frac{x^2}{x^2} - \frac{4}{x^2}}} = \lim_{x \rightarrow -\infty} \frac{\frac{1}{x} + 1}{-\sqrt{1 - \frac{4}{x^2}}} = \frac{0+1}{-\sqrt{1-0}} = -1$ $\therefore$ The horizontal asymptotes are $y=1$ and $y=-1$.	K1 K1 J1J1

5.(b)(i)	$\lim_{x \rightarrow -2^-} 1 - ax^2 = \lim_{x \rightarrow -2^+} 2x + a$ $1 - 4a = 2(-2) + a$ $a = 1$	K1 K1 J1
5.(b)(ii)	$\lim_{x \rightarrow -2^+} \frac{2x+1-(-3)}{x+2} = \lim_{x \rightarrow -2^+} \frac{2(x+2)}{x+2} = \lim_{x \rightarrow -2^+} 2 = 2$ $\lim_{x \rightarrow -2^-} \frac{1-x^2-(-3)}{x+2} = \lim_{x \rightarrow -2^-} \frac{(2+x)(2-x)}{x+2} = \lim_{x \rightarrow -2^+} (2-x) = 4$ Since $\lim_{x \rightarrow -2^-} \frac{f(x)-f(-2)}{x+2} \neq \lim_{x \rightarrow -2^+} \frac{f(x)-f(-2)}{x+2}$ $\therefore$ The limit does not exist.	K1 K1 K1 J1
TOTAL		14 Marks

NO	ANSWER SCHEME	MARKS
6 (a)(i)	$x = e^{2t+1}, \quad y = e^{-(2t-1)}.$ $\frac{dx}{dt} = 2e^{2t+1}, \quad \frac{dy}{dt} = -2e^{-(2t-1)}$ $\frac{dy}{dx} = -2e^{-(2t-1)} \cdot \frac{1}{2e^{2t+1}}$ $= -e^{-(2t-1)} \cdot e^{-(2t+1)}$ $= -e^{-4t}$ $\frac{d^2y}{dx^2} = 4e^{-4t} \cdot \frac{1}{2e^{2t+1}}$ $= 2e^{-6t-1}$ at (e^3, e^{-1}), $x = e^3 \Rightarrow e^3 = e^{2t+1} \Rightarrow t = 1$ $\left. \frac{dy}{dx} \right _{t=1} = -e^{-4(1)}$ $= -e^{-4}$ $\left. \frac{d^2y}{dx^2} \right _{t=1} = 2e^{-6(1)-1}$ $= 2e^{-7}$	B1 K1 K1 K1 (sub both) J1 J1
(a)(ii)		K1 J1

(b)	$z = x^2 - xy$ $= (e^{2t+1})^2 - (e^{2t+1})(e^{-(2t-1)})$ $= e^{4t+2} - e^2$ $\frac{dz}{dt} = 4e^{4t+2}$ $\frac{dz}{dt} > 0 \Rightarrow 4e^{4t+2} > 0$ $\therefore \{t : t \in \mathbb{R}\}$ $y = \tan\left(\frac{\pi}{4} + x\right)$ $\frac{dy}{dx} = \sec^2\left(\frac{\pi}{4} + x\right)$ $\frac{d^2y}{dx^2} = 2\sec\left(\frac{\pi}{4} + x\right)\left(\sec\left(\frac{\pi}{4} + x\right)\tan\left(\frac{\pi}{4} + x\right)\right)$ $\frac{d^2y}{dx^2} = 2\sec^2\left(\frac{\pi}{4} + x\right)\tan\left(\frac{\pi}{4} + x\right)$ $\frac{d^2y}{dx^2} = 2\left(\frac{dy}{dx}\right)(y)$ $\frac{d^2y}{dx^2} - 2y\left(\frac{dy}{dx}\right) = 0 \text{ (shown)}$	J1 K1 J1 K1 K1 K1 (sub $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$) J1
TOTAL		15 Marks
7(a)	$V = \pi\left(5h^2 - \frac{1}{3}h^3\right)$ $\frac{dV}{dt} = \pi(10h - h^2)\frac{dh}{dt}$ when $h = 2$, $\frac{dV}{dt} = 4\pi$ $4\pi = \pi(20 - 2^2)\frac{dh}{dt}$ $\frac{dh}{dt} = \frac{1}{4} \text{ cm/s}$	K1 differentiate K1 substitute J1

7(b)	$r^2 + (5-h)^2 = 5^2$ $2r \frac{dr}{dt} + 2(5-h) \left(-\frac{dh}{dt} \right) = 0$ $r \frac{dr}{dt} - (5-h) \left(\frac{dh}{dt} \right) = 0$ when $h = 2$, $r^2 + (5-2)^2 = 5^2$ $r^2 + 9 = 25$ $r^2 = 16$ $r = \pm 4$ $(r > 0), r = 4$ $4 \frac{dr}{dt} - (5-2) \left(\frac{1}{4} \right) = 0$ $4 \frac{dr}{dt} - \frac{3}{4} = 0$ $4 \frac{dr}{dt} = \frac{3}{4}$ $\frac{dr}{dt} = \frac{3}{16} \text{ cm/s}$	K1 – r in terms of h K1 – differentiate K1 – substitute K1 – ± 4 J1 – value of r K1 – substitute J1 – answer
	TOTAL	10 Marks
	SECTION B	75 Marks